

LECTURE 2 · WEB TECHNOLOGIES

Introduction to HTML

Md Istakiak Adnan Palash
Lecturer, Dept. of ICT
Bangladesh University of Professionals (BUP)

HTML

- HTML stands for HyperText Markup Language. HTML is the basic building block of World Wide Web
- HTML is the standard markup language for creating Web pages
- HTML describes the structure of a Web page
- HTML consists of a series of elements
- HTML elements tell the browser how to display the content

HTML Page Structure

```
<html>
  <head>
    <title>Page title</title>
  </head>
  <body>
    <h1>This is a heading</h1>
    <p>This is a paragraph.</p>
    <p>This is another paragraph.</p>
  </body>
</html>
```

Basic Structure of HTML

- HTML tags are mark up codes used for formatting ASCII (text) files, which is interpreted by the browser when displaying a web page.
- All HTML documents must start with a document type declaration: **<!DOCTYPE html>.**
- HTML **tags** basically include the tag name enclosed by angle brackets (< >). for example, <HTML>

Basic Structure of HTML

- Most tags are used in pairs, a **begin tag** and an **end tag**, for example <title>…</title>. The back slash on the tag signifies it as the end tag. Some tags do not require end tags.
- In coding HTML tags, you can use capital letters, lower case, or a combination of both. For easy editing it is recommended that you use lower case letters for most purposes and capital letters for special cases.

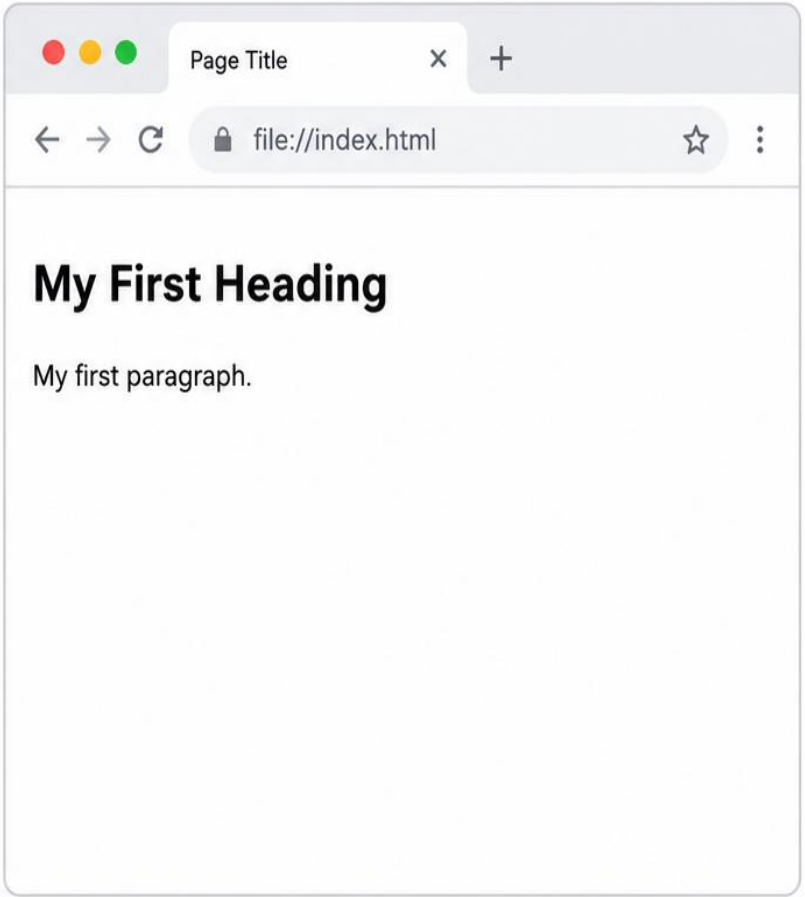
A Simple HTML Document

index.html

```
1  <!DOCTYPE html>
2  <html>
3    <head>
4      <title>Page Title</title>
5    </head>
6    <body>
7
8      <h1>My First Heading</h1>
9
10     <p>My first paragraph.</p>
11
12   </body>
14 </html>
```

Head

Body



Example Explained

- The <!DOCTYPE html> declaration defines that this document is an HTML document
- The <html> element is the root element of an HTML page
- The <head> element contains meta information about the HTML page
- The <title> element specifies a title for the HTML page (which is shown in the browser's title bar or in the page's tab)
- The <body> element defines the document's body, and is a container for all the visible contents, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.
- The <h1> element defines a large heading
- The <p> element defines a paragraph

Elements

- HTML documents are defined by HTML elements.
- An HTML element is everything between the start tag and the end tag.
- The start tag is often called the opening tag.
- The end tag is often called the closing tag.
- All the arguments and values within an opening and closing tag are called an element.
- An element may include attributes, values, object, reference etc.
- There are many types of elements are in the body of a Web documents such as ***block-level elements, Inline elements*** etc.

Attribute and Values

- **Attribute**
  - The attribute further defines the properties of the HTML tag.
  - For example, the **<table>** tag has several attributes including align, border, height, width, etc.

- **Values**
  - A value is a string that provides description or characteristics of an element or attribute of an element.

HTML Headings

- Headings help in defining the hierarchy and the structure of the web page content.

- HTML offers six levels of heading tags, **<h1>** through **<h6>;** the lower the heading level number, the greater its importance; therefore <h1> tag defines the most important heading, whereas the <h6> tag defines the least important heading in the document.

Headings Are Important

- Search engines use the headings to index the structure and content of your web pages.
- Users often skim a page by its headings. It is important to use headings to show the document structure.
- <h1> headings should be used for main headings, followed by <h2> headings, then the less important <h3>, and so on.

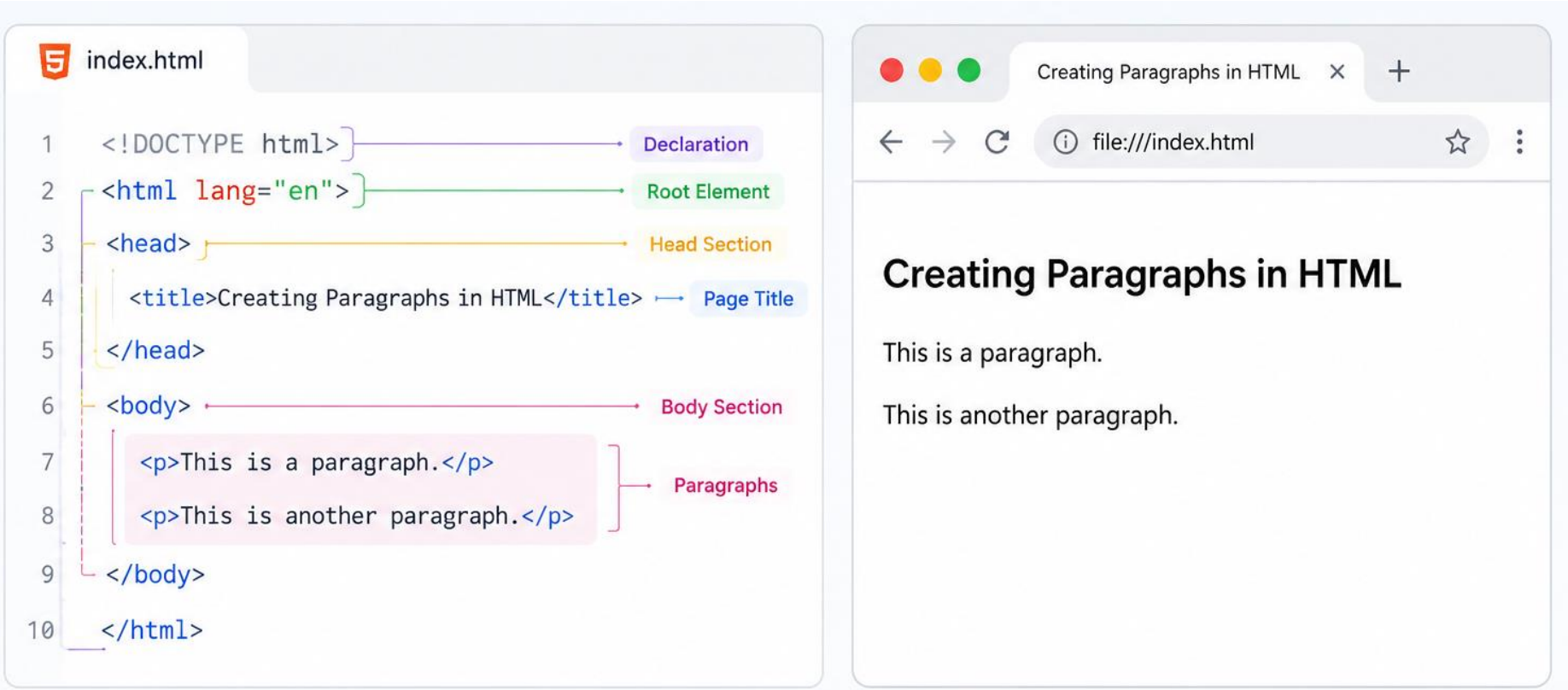
For example:

- <h1> - Page title
- <h2> - Section titles
- <h3> - Sub-sections

HTML Paragraphs

- The HTML <p> element defines a paragraph.

- A paragraph always starts on a new line, and browsers automatically add some white space (a margin) before and after a paragraph.


index.html
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <title>Creating Paragraphs in HTML</title>
5 </head>
6 <body>
7 This is a paragraph.
8 This is another paragraph.
9 </body>
10 </html>
Declaration
Root Element
Head Section
Page Title
Body Section
Paragraphs
Creating Paragraphs in HTML
file:///index.html
Creating Paragraphs in HTML
This is a paragraph.
This is another paragraph.

< br / > tag

- Line Breaks are given Using the < br / > tag
- The line break element specifies that a new line must be started at the given point.
- Example:

```
<p>
Mary had a little lamb< br />
It's fleece was white as snow< br />
Everywhere that Mary went< br />
She was followed by a little lamb.
</p>
```

<hr> tag

- Insert a horizontal line
- A horizontal rule element is a divider between sections of text such as a full width horizontal rule or equivalent graphic.
- You can use the <hr> tag to create horizontal rules or lines to visually separate content sections on a web page.
- Like
, the <hr> tag is also an empty element.

The Poem Problem

- This poem will display on a single line:

Twinkle, twinkle, little star,
How I wonder what you are!
Up above the world so high,
Like a diamond in the sky.

Solution - The HTML <pre> Element

- The HTML <pre> element defines preformatted text.
- The text inside a <pre> element is displayed in a fixed-width font (usually Courier), and it preserves both spaces and line breaks:

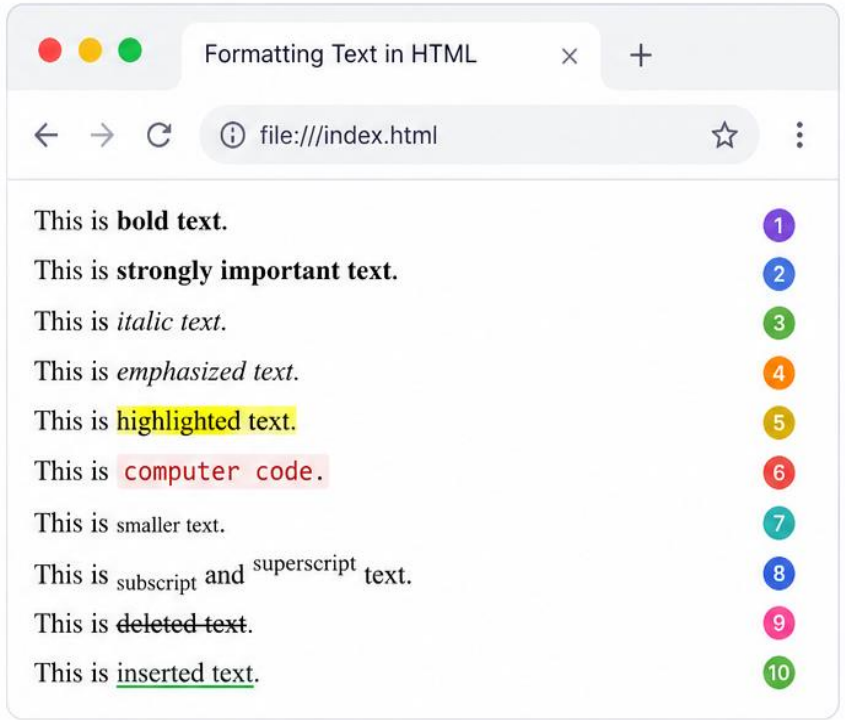
```
<pre>
    Twinkle, twinkle, little star,
    How I wonder what you are!
    Up above the world so high,
    Like a diamond in the sky.
</pre>
```

HTML Text Formatting

- HTML provides several tags that you can use to make some text on your web pages to appear differently than normal text.For example, you can use the
  - tag **<b>** to make the text bold
  - tag **<i>** to make the text italic
  - tag **<u>** to underline the text
  - tag **<mark>** to highlight the text
  - tag **<code>** to display a fragment of computer code
  - tags **<ins>** and **<del>** for marking editorial insertions and deletions

index.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
   <title>Formatting Text in HTML</title>
</head>
<body>
   <p>This is <b>bold text</b>.</p>
   <p>This is <strong>strongly important text</strong>.</p>
   <p>This is <i>italic text</i>.</p>
   <p>This is <em>emphasized text</em>.</p>
   <p>This is <mark>highlighted text</mark>.</p>
   <p>This is <code>computer code</code>.</p>
   <p>This is <small>smaller text</small>.</p>
   <p>This is <sub>subscript</sub> and <sup>superscript</sup> text.</p>
   <p>This is <del>deleted text</del>.</p>
   <p>This is <ins>inserted text</ins>.</p>
</body>
</html>
```



Difference between and <b> tag

- Both **** and **<b>** tags render the enclosed text in a bold typeface by default, but the **** tag indicates that its contents have strong importance, whereas the **<b>** tag is simply used to draw the reader's attention without conveying any special importance.
- **Example:**

<*p*><*strong*>WARNING! Please proceed with caution.</*p*>

<*p*>The concert will be held at <*b*>Hyde Park</*b*> in London.</*p*>

Difference between and <i> tag

- Both **** and **<i>** tags render the enclosed text in italic type by default, but the **** tag indicates that its contents have stressed emphasis compared to surrounding text, whereas the **<i>** tag is used for marking up text that is set off from the normal text for readability reasons, such as a technical term, an idiomatic phrase from another language, a thought, etc.
- **Example:**

<*p*>Cats are <*em*>cute</*em*> animals.</*p*>

<*p*>The <*i*>Royal Cruise</*i*> sailed last night.</*p*>

Formatting Quotations

- You can easily format the quotation blocks from other sources with the HTML **<blockquote>** tag.
- Blockquotes are generally displayed with indented left and right margins, along with a little extra space added above and below.
- **Example:**

<*blockquote*>

<*p*>Learn from yesterday, live for today, hope for tomorrow. The important thing is not to stop questioning.</*p*>

<*cite*>— Albert Einstein</*cite*>

</*blockquote*>

Showing Abbreviations

- An abbreviation is a shortened form of a word, phrase, or name.
- You can use the <abbr> tag to denote an abbreviation.
- The **title** attribute is used inside this tag to provide the full expansion of the abbreviation.
- **Example:**

The <abbr title="World Wide Web Consortium">W3C</abbr> is the main international standards organization for the <abbr title="World Wide Web">WWW or W3</abbr>. It was was founded by Tim Berners-Lee.

HTML Links

- A link or hyperlink is a connection from one web resource to another.
- Links allow users to move seamlessly from one page to another, on any server anywhere in the world.
- A link has two ends, called anchors. The link starts at the source anchor and points to the destination anchor, which may be any web resource, for example, an image, an audio or video clip, a PDF file, an HTML document or an element within the document itself, and so on.

HTML Link Syntax

- Links are specified in HTML using the <a> tag.
- A link or hyperlink could be a word, group of words, or image.

```
<a href="url">Link text</a>
```

- **Example:**

```
<a href="https://www.google.com/">Google Search</a>
<a href="https://www.bup.edu.bd/">BUP</a>
<a href="images/kites.jpg">
   <img src="kites-thumb.jpg" alt="kites">
</a>
```

Setting the Targets for Links

- The **target** attribute tells the browser where to open the linked document. There are four defined targets, and each target name starts with an underscore(_) character:
  - **_blank —** Opens the linked document in a new window or tab.
  - **_parent —** Opens the linked document in the parent window.
  - **_self —** Opens the linked document in the same window or tab as the source document. This is the default, hence it is not necessary to explicitly specify this value.
  - **_top —** Opens the linked document in the full browser window.

Image Handling in HTML

- The Image element **<img….>** is used to incorporate in-line graphics (typically icons, images) into an HTML document.
- Browsers that cannot render in-line images ignore the Image element unless it contains the ALT attribute.
- The <img> tag is empty, which means that it contains attributes only, and has no closing tag.

<*img src*="url" *alt*="some_text">

- **Example:**

<*img src*="kites.jpg" *alt*="Flying Kites">

Attributes of image element

Image Element Attributes	Description
SRC	Specify the URL of the image to be displayed. Example: src="photo.jpg"
ALT	Specify an alternate text for an image/video, if the image cannot be displayed. Example: alt="Sunset"
WIDTH	Define the width of a graphics element. Example: width="500"
HEIGHT	Define the height of a graphics element. Example: height="600"
ALIGN	Specify the alignment of images. Example: align="center" ⓘ Deprecated in HTML5. Use CSS instead.
BORDER	Control the thickness of the border around an image displayed. Example: border="2" ⓘ Deprecated in HTML5. Use CSS instead.

tag

- The tag defines a division or a section in an HTML document.
- The tag is used to group block-elements to format them with styles.
- The element is very often used together with CSS, to layout a web page.
- By default, browsers always place a line break before and after the element. However, this can be changed with CSS.
- **Example:**

```
<div style="color:#00FF00">
<h3>This is a header</h3>
<p>This is a paragraph.</p>
</div>
```

tag

- The tag is used to group inline-elements in a document.
- The tag provides no visual change by itself. The tag provides a way to add a hook to a part of a text or a part of a document.
- When the text is hooked in a element you can add styles to the content, or manipulate the content with for example JavaScript.
- It does not add any new line.
- Example

```
<p>
My mother has <span style="color:lightblue;font-weight:bold">light
blue</span>
eyes and my father has <span style="color:darkolivegreen;font-
weight:bold">dark green</span> eyes.
</p>
```

Thank You!